Sahaj Bhadja

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Summary

Software Engineer with 2+ years shipping production **AI** and **full-stack** features, from **LLM workflows** that automate paperwork-heavy evaluation to a hackathon-winning **healthcare** app with health-record APIs, building AI-native with Claude daily.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed a multi-tenant platform on Azure SQL and Blob Storage with **row-level security** isolating sensitive records for 500+ users, cutting dashboard latency from 1.2s to 350ms via query and indexing work.
- Built fault-tolerant **LLM workflows** with FastAPI background workers and explicit job states, processing 1K+ AI jobs at 3x throughput while integrations ran 5K+ monthly jobs with validation at 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Automated paperwork-heavy grading with **LLM pipelines** that generate rubrics, read handwritten answers via OCR, and return feedback using GPT-4o, Claude, and Qwen-VL, cutting turnaround 60% for 150+ instructors.
- Built React views and Flask **REST APIs** for uploads and grading history with secure PDF, DOCX, and ZIP ingestion, handling 10K+ daily requests and 1K+ files per batch with fault-tolerant execution.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Wrote 20+ **unit and integration test** suites with PyTest and Mockito and automated nightly Jenkins runs across Windows, Linux, and macOS, generating coverage reports and failure logs in Agile sprints.
- Debugged and optimized large-scale API integrations with pagination, field filtering, and caching, improving response times 30% for low-bandwidth users, and led an ETL redesign that boosted **pipeline efficiency** 68%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a MERN resource-management platform with eight microservices for authentication, resource allocation, billing, and email and SMS notifications, syncing 2M+ records across locations via REST APIs and real-time **WebSockets**.
- Automated Jenkins **CI/CD** with unit, integration, and end-to-end tests using Jest, Mocha, and Selenium, reaching 95%+ build validation, and used usage analytics to cut resource wastage 42%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Translated **client requirements** into an end-to-end booking architecture with 12+ integration points, then built Python Flask APIs for bookings, pricing, and coupons with Firebase real-time status updates.
- Automated refund and dispute handling through idempotent payment **webhooks**, cutting manual work 70%, and used Firebase transactional locking to prevent double-booked time slots across 50K+ booking records.

Skills

Languages & Fundamentals: Python, TypeScript, JavaScript, SQL, Java, C++, Data Structures, Algorithms, Operating Systems
Full Stack: React, Next.js, Node.js, FastAPI, Flask, REST APIs, WebSockets, TailwindCSS
AI: Claude, Claude Code, GPT-4o, LLM Workflows, Prompt Chaining, RAG, OCR
Healthcare & Integrations: Health-Record APIs (ABDM), Consent-Based Data Access, Twilio SMS, Webhooks
Data & Cloud: PostgreSQL, Supabase, MongoDB, Row-Level Security, Azure, AWS, Docker, CI/CD
Testing & Practices: Unit Testing, Integration Testing, PyTest, Code Reviews, Debugging

Projects

HMIS – Hospital Management Information System Live | *TypeScript, MERN, ABDM APIs, Twilio, Sentry*

- Won Smart India Hackathon building a healthcare app with consent-based **health-record APIs** (ABDM), appointment workflows, and Twilio patient SMS reminders, giving doctors 2x faster history access across 100+ daily appointments.

Big-Brain Live | *LLaMA-2, FAISS, Clerk, Convex, PostgreSQL*

- Built a document assistant that summarizes long files and answers questions with LLaMA-2 **RAG** over FAISS, cutting document review time 60% for 200+ users with Clerk-secured access.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel-queue **scheduling algorithms** in an interactive React simulator with Gantt charts, testing 1K+ process combinations and demonstrating up to 25% lower average turnaround.